

The officials from the Bhutanese government, DGPC and AGEL after signing of the MoU in Thimphu on May 8.

referred to as the "hermit kingdom".

The King has turned its gaze on the nation's all-weather friend India's leading infrastructure company, the Adani Group.

In July 2024, the King and Bhutan's Prime Minister (PM) Tshering Tobgay visited Adani Green Energy's 30-gigawatt (GW) renewable energy site at Khavda and Mundra Port in Gujarat.

On that trip, the King and PM Tobgay met Adani Group's Chairman Gautam Adani in Ahmedabad. The meeting proved to be a gamechanger, leading to a growing trust between the Bhutanese government and the Adani Group.

And 10 months on, a landmark Memorandum of Understanding (MoU) was signed in Thimphu on May 8. The MoU was signed between Adani Green Energy Limited (AGEL) and Bhutan's Druk Green Power Corporation (DGPC) to jointly develop 5,000 megawatt (MW) of hydro power projects in the South Asian nation.

The MoU was signed in Thimphu by AGEL's Chief Operating Officer (COO), Pump Storage Projects (PSP) and Hydro, Naresh Telgu and DGPC's Managing Director (MD) Dasho Chhewang Rinzin. PM Tobgay, Bhutan's Minister for Energy and Natural Resources, Lyonpo Gem Tshering, and other dignitaries were also present on the occasion.

This MoU builds on the ongoing partnership for the 570 MW peaking run-of-the-river Wangchhu Hydropower Project (HPP). The broader 5,000 MW initiative will encompass additional hydropower and PSPs to be identified over the course of time. Detailed Project Reports (DPRs) have been prepared and taken forward for implementation in phases.

The Wangchhu HPP can be designed either as a water storage scheme or as a peaking run-of-the-river scheme. According to experts, the former requires a large reservoir that submerges a significant amount of land, impacting nearby forests and residential areas considerably, leading to resettlement and rehabilitation, along with increased project costs. In contrast, peaking run-of-the-river projects require a much smaller pondage area, which reduces land requirements and forest submergence, making them more sustainable and commercially viable, the experts maintain.

Wangchhu HPP is on a fast track

The Adani Group is looking at signing the shareholders' agreement (SHA) for Wangchhu HPP shortly.

Once the SHA is signed, up to 10 months will be required for pre-development activities, supplementary investigations, design optimization and award of Civil, Electrical and Mechanical (E&M) contracts for this project. Immediately after the award of contract, project construction activities will start, and it is likely to take around 60 months for commissioning and start of operation of the project.

A win-win collaboration

The DGPC, Bhutan's premier hydropower developer, has decades of experience in managing the nation's renewable energy resources. It plays a pivotal role in Bhutan's clean energy journey, contributing to both domestic energy security and sustainable development. Through such partnerships, the DGPC is also helping to strengthen Bhutan's position in regional energy cooperation.

As part of this collaboration, the Adani Group will ensure reliable power offtake and integration with India's commercial power markets, further reinforcing Bhutan's role in the regional energy trade. The partnership is strongly backed by the Royal Government of Bhutan and the Government of India, underscoring a shared vision for clean energy growth and economic integration.

This initiative also aligns with Bhutan's renewable energy roadmap, which aims to achieve an additional 20,000 MW of generation capacity by 2040. The roadmap prioritises diversification into solar and geothermal energy and encourages strategic partnerships to attract investment and innovation.

AGEL's dream run

May 2022: AGEL's Hydro and PSP business unit starts operation

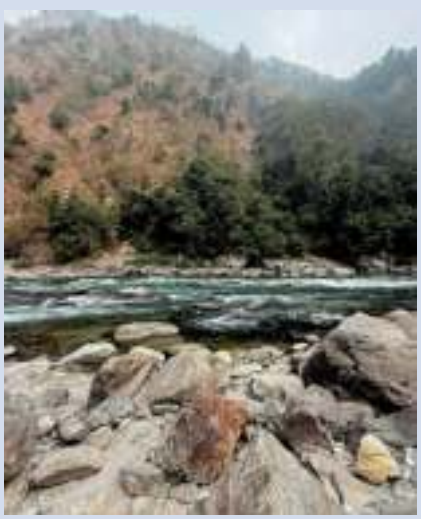
May 2022 – June 2025: 4,800 MW PSPs are under construction, including 500 MW MW in Chitravathi, Andhra Pradesh (AP); 2,019 MW in Gandikota also in AP; and 1,695 MW in Tarali, Maharashtra.

Hydel haven

Total technical potential: 37,000 MW

Techno-commercially feasible potential: 33,000 MW

Installed capacity: 2,344 MW, which is around 26% of Delhi's peak demand of around 9,000 MW this summer.



The Wangchhu HPP site, which is located around 130 km from the capital Thimphu.